[Abstract]

Attention Is All You Need

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Abstract: The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely.

Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU.

[Model Architecture & Scaled Dot-Product Attention]

3.2 Attention: An attention function can be described as mapping a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. We call our particular attention Scaled Dot-Product Attention. The input consists of queries and keys of dimension d_k, and values of dimension d_v.

We compute the dot products of the query with all keys, divide each by the square root of d_k (1 / sqrt(d_k) scaling factor), and apply a softmax function to obtain the weights on the values. For large values of d_k, the dot products grow large in magnitude, pushing the softmax function into regions where it has extremely small gradients.

Multi-Head Attention allows the model to jointly attend to information from different representation subspaces at different positions. With a single attention head, averaging inhibits this. In this work we employ h = 8 parallel attention layers, or heads, with d_k = d_v = d_model / h = 64.

[Positional Encoding & Experimental Results]

3.5 Positional Encoding: Since our model contains no recurrence and no convolution, in order for the model to make use of the order of the sequence, we must inject some information about the relative or absolute position of the tokens in the sequence. We use sine and cosine functions of different frequencies added to the input embeddings.

5.1 Machine Translation Results: On the WMT 2014 English-to-German translation task, the big Transformer model (Transformer big) outperforms the best previously reported models including ensembles by more than 2.0 BLEU, establishing a new state-of-the-art BLEU score of 28.4 after training for 3.5 days on 8 P100 GPUs.

On the WMT 2014 English-to-French translation task, our big model achieves a BLEU score of 41.8, outperforming all of the previously published single models, at less than 1/4 the training cost of the previous state-of-the-art model.